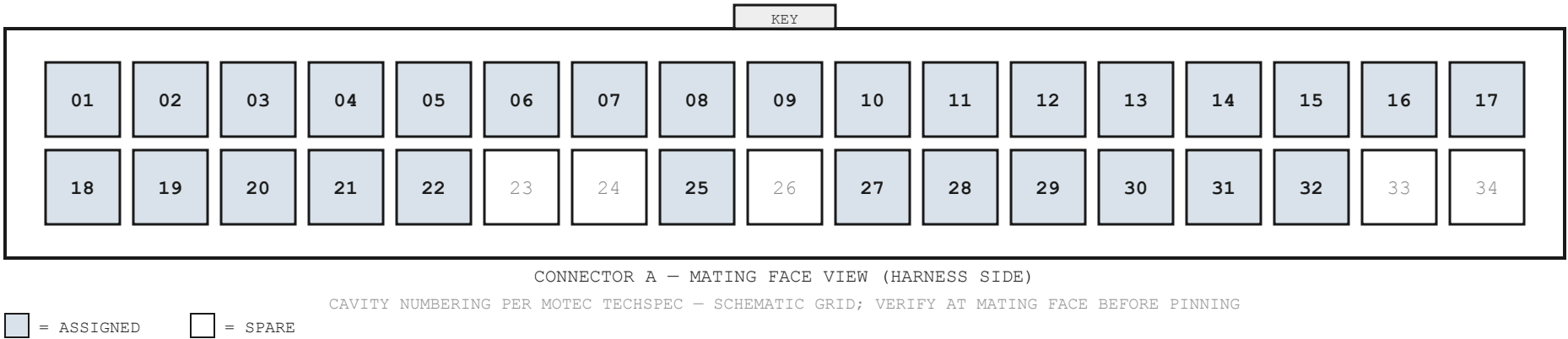


M130 ECU — CONNECTOR A (34-WAY TYCO SUPERSEAL)

1977 CHEVROLET K5 BLAZER — LS3 6.2L — MoTeC M130 ENGINE HARNESS



PIN	M130 FUNCTION	WIRE #	CIRCUIT	COLOR	AWG	DESTINATION
A01	OUT_HB2	#4a	ETB TAC Motor 2	BRN	20	GM 12605109 Pin A (throttle body)
A02	SEN_5V_A	#4e, 99r, 102r, 112r	5V reference rail A	GRY	22	ETB Pin E + CKP + OPS + Fuel Press 5V supplies
A03	IGN_LS1	#5	Ignition Coil 3	WHT	22	DEL-Stributor central mount
A04	IGN_LS2	#7	Ignition Coil 5	WHT/WHT	22	DEL-Stributor central mount
A05	IGN_LS3	#8	Ignition Coil 7	WHT/BLK	22	DEL-Stributor central mount
A06	IGN_LS4	#9	Ignition Coil 2	WHT/RED	22	DEL-Stributor central mount
A07	IGN_LS5	#10	Ignition Coil 4	WHT/BLU	22	DEL-Stributor central mount
A08	IGN_LS6	#11	Ignition Coil 6	WHT/YEL	22	DEL-Stributor central mount
A09	SEN_5V_B	#108r, 101r	5V reference rail B	GRY	22	MAP + CMP 5V supplies
A10	BAT_NEG1	—	Battery negative	—	—	Sensor star ground (appx G \$2.3)
A11	BAT_NEG2	—	Battery negative	—	—	Sensor star ground (appx G \$2.3)
A12	IGN_LS7	#12	Ignition Coil 8	WHT/VIO	22	DEL-Stributor central mount
A13	IGN_LS8	#24	Ignition Coil 1	WHT/ORG	22	DEL-Stributor central mount
A14	AV1 RESOLVED v4: AV1 collision cleared — #102 Oil Press moved to A25/AV5; ETB keeps A14 (newer decision wins). 2026-04-13 harness-spec AV map = STALE	#4c	ETB TPS1 Signal	DK GRN	22	GM 12605109 Pin C (throttle body)
A15	AV2 v4: stays AV2 — 2026-04-13 harness-spec AV map STALE	#108	MAP Sensor	VIO/WHT	22	ACDelco PT2293 / Bosch 3-pin
A16	AV3 RESOLVED v4: stays AV3 — 2026-04-13 harness-spec AV map (Oil Press on AV3) marked STALE	#112	Fuel Pressure Sensor	VIO/BLK	22	Fuel rail pressure sender
A17	AV4 RESOLVED v4: stays AV4 — 2026-04-13 harness-spec AV map (TPS1 on AV4) marked STALE	#4d	ETB TPS2 Signal	PPL	22	GM 12605109 Pin D (throttle body)
A18	OUT_HB1	#4b	ETB TAC Motor 1	YEL	20	GM 12605109 Pin B (throttle body)
A19	INJ_PH1	#13	Fuel Injector 1	GRN	22	EV6/USCAR @ fuel rail cyl 1
A20	INJ_PH2	#14	Fuel Injector 2	GRN/WHT	22	EV6/USCAR @ fuel rail cyl 2
A21	INJ_PH3	#15	Fuel Injector 3	GRN/BLK	22	EV6/USCAR @ fuel rail cyl 3
A22	INJ_PH4	#16	Fuel Injector 4	GRN/RED	22	EV6/USCAR @ fuel rail cyl 4
A23	— SPARE —	—	—	—	—	—
A24	— SPARE —	—	—	—	—	—
A25	AV5 RESOLVED v4: #102 MOVED A14 → A25 (AV5) per receipts/2026-06-09_cut-list-v4.md — only free AV pin; zero recrimps elsewhere	#102	Oil Pressure (ECU)	VIO	22	Oil press sender (galley); gnd #102g → B16, 5V #102r → A02
A26	— SPARE —	—	—	—	—	—
A27	INJ_PH5	#17	Fuel Injector 5	GRN/BLU	22	EV6/USCAR @ fuel rail cyl 5
A28	INJ_PH6	#18	Fuel Injector 6	GRN/YEL	22	EV6/USCAR @ fuel rail cyl 6
A29	INJ_PH7	#19	Fuel Injector 7	GRN/VIO	22	EV6/USCAR @ fuel rail cyl 7
A30	INJ_PH8	#20	Fuel Injector 8	GRN/ORG	22	EV6/USCAR @ fuel rail cyl 8
A31	OUT_HB3 v4: spare half-bridge as tach pulse (resolved 2026-05-14, K5_WIRING_STATE.md \$3); requires M1 GPR config at tune	#116	Dakota Tach Signal	WHT/BLK	22	Dakota VHX cluster tach input (dash)
A32	OUT_HB4 v4: spare half-bridge as VSS mirror (resolved 2026-05-14, K5_WIRING_STATE.md \$3); requires M1 GPR config at tune	#118	Dakota VSS / Speedo	WHT/YEL	22	Dakota VHX cluster speedo input (dash)
A33	— SPARE —	—	—	—	—	—
A34	— SPARE —	—	—	—	—	—

NOTES:

Cut-list wires with FROM = 'ECU' (switch/digital inputs #33,36,37,39-46,53,98,100,105-107,111) have no M130 pin in cut list v4 — pin assignment TBD.

#98 Fuel Level Sender needs an AV pin: 2026-05-14 Dakota addendum claimed A25/AV5, but v4 assigns A25 to #102 Oil Press — next free AV input per MoTeC techspec, or Skylar call (v4 open item 1).

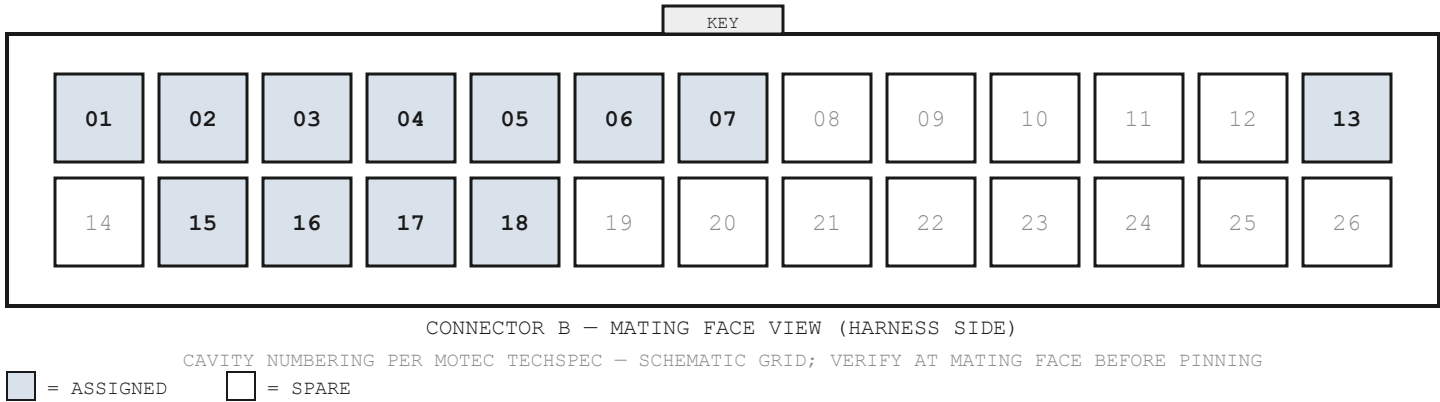
Sources: pdm_power_budget.md (M130 pins in use) + K5_cut_list_v4.txt (wires, via FROM column) + appendix-g (BAT_NEG / SEN_OV / CAN pins) + receipts/2026-06-09_cut-list-v4.md (AV shuffle, Dakota A31/A32).

Spare cavities carry no terminal; plug with sealed cavity plugs per Superseal practice.

NUKE LTD / DESERT PERFORMANCE
1977 CHEVROLET K5 BLAZER — VIN CKR187F127263
LS3 6.2L / MoTeC M130 + PDM30
SHEET 1 OF 3 — 2026-06-09 — REV A
DERIVED FROM K5_cut_list_v4 + pdm30_channel_plan; lengths/land marks per calc engine

M130 ECU — CONNECTOR B (26-WAY TYCO SUPERSEAL)

1977 CHEVROLET K5 BLAZER — LS3 6.2L — MoTeC M130 ENGINE HARNESS



PIN	M130 FUNCTION	WIRE #	CIRCUIT	COLOR	AWG	DESTINATION
B01	UDIG1	#99	Crank Position Sensor	BLU/WHT/WHT	22	3-pin Metri-Pack CKP; SHLD 2C, drain → B15 (#99s)
B02	UDIG2 RESOLVED v4: Rear Backup Camera on B02 (2026-04-13 spec) marked STALE — B02 = CMP; camera pwr = PDM30:OUT15 (#97)	#101	Cam Position Sensor	BLU/WHT/BLK	22	CMP; SHLD 2C, drain → B16 (#101s)
B03	AT1 CONFLICT: pdm_power_budget M130 list swaps CLT/IAT on B03/B04; cut list v4 + appendix-g value shown	#109	Intake Air Temp Sensor	TAN	22	IAT; gnd #109g → B15
B04	AT2 CONFLICT: see B03 — cut list v4 + appendix-g value shown	#110	Coolant Temp Sensor (ECU)	TAN/WHT	22	CLT; gnd #110g → B16
B05	AT3	#113	Oil Temperature Sensor	TAN/BLK	22	OTS; gnd #113g → B15
B06	AT4 SPARE AT4 — no signal wire yet (cut list v4 KNOWN GAPS); companion gnd needed when assigned	—	(reserved — no wire)	—	—	—
B07	KNOCK1	#103	Knock Sensor Bank 1	GRY	22	Piezo 2-pin; SHLD 2C, drain → B15
B08	— SPARE —	—	—	—	—	—
B09	— SPARE —	—	—	—	—	—
B10	— SPARE —	—	—	—	—	—
B11	— SPARE —	—	—	—	—	—
B12	— SPARE —	—	—	—	—	—
B13	KNOCK2	#104	Knock Sensor Bank 2	GRY/WHT	22	Piezo 2-pin; SHLD 2C, drain → B16
B14	— SPARE —	—	—	—	—	—
B15	SEN_OV_A	#4f, 99g, 99s, 103g, 103s, 108g, 109g, 113g	Sensor OV star A	BLK / BARE	22	ETB + CKP + Knock1 gnds/drains, MAP + IAT + OTS grounds
B16	SEN_OV_B	#101g, 101s, 102g, 104g, 104s, 110g, 112g	Sensor OV star B	BLK / BARE	22	CMP + Knock2 gnds/drains, OPS + CLT + FuelPress grounds
B17	CAN_H v4: #125 Holley T43 CAN stub (2.0ft) splices trunk #62 — stub is 22 AWG vs 24 AWG trunk, flagged in v4 receipt	#62	CAN Bus Network	WHT/GRN	24	PDM30 B25 + C125/Dakota + T43 TCU (#125 stub); 120Ω ea end
B18	CAN_L	#62	CAN Bus Network	WHT/GRN	24	PDM30 B26 + C125/Dakota + T43 TCU (#125 stub); 120Ω ea end
B19	— SPARE —	—	—	—	—	—
B20	— SPARE —	—	—	—	—	—
B21	— SPARE —	—	—	—	—	—
B22	— SPARE —	—	—	—	—	—
B23	— SPARE —	—	—	—	—	—
B24	— SPARE —	—	—	—	—	—
B25	— SPARE —	—	—	—	—	—
B26	— SPARE —	—	—	—	—	—

NOTES:

Cut-list wires with FROM = 'ECU' (switch/digital inputs #33,36,37,39-46,53,98,100,105-107,111) have no M130 pin in cut list v4 — pin assignment TBD.

#98 Fuel Level Sender needs an AV pin: 2026-05-14 Dakota addendum claimed A25/AV5, but v4 assigns A25 to #102 Oil Press — next free AV input per MoTeC techspec, or Skylar call (v4 open item 1).

Sources: pdm_power_budget.md (M130 pins in use) + K5_cut_list_v4.txt (wires, via FROM column) + appendix-g (BAT_NEG / SEN_OV / CAN pins) + receipts/2026-06-09_cut-list-v4.md (AV shuffle, Dakota A31/A32).

Spare cavities carry no terminal; plug with sealed cavity plugs per Superseal practice.

NUKE LTD / DESERT PERFORMANCE
1977 CHEVROLET K5 BLAZER — VIN CKR187F127263
LS3 6.2L / MoTeC M130 + PDM30
SHEET 2 OF 3 — 2026-06-09 — REV A
DERIVED FROM K5_cut_list_v4 + pdm30_channel_plan; lengths/land marks per calc engine

PDM30 — OUTPUT CHANNELS

1977 CHEVROLET K5 BLAZER — LS3 6.2L — MoTeC PDM30 POWER DISTRIBUTION
CHANNEL PLAN: K5_pdm30_channel_plan.md (2026-04-05, AUTHORITATIVE) — 30/30 CHANNELS USED, 0 SPARE

CH	RATING	DEVICE / GROUP	EST AMPS	WIRE #	COLOR	AWG
OUT1	20A	Radiator Fan 1	18A	#21	ORN/WHT	12
OUT2	20A	Radiator Fan 2	18A	#22	ORN/BLK	12
OUT3	20A	Power Window Motor Left	15A	#34	DK BLU	16
OUT4	20A	Power Window Motor Right	15A	#35	DK BLU/WHT	16
OUT5	20A	Electric Water Pump	12A	#25	ORN/BLU	14
OUT6	20A	Heater Blower Motor	12A	#51	ORN/BLK/BLU	10
OUT7	20A v4: #126 dash	Electric Parking Brake (E-Stopp) button trigger (22 AWG ORN/WHT, 10.0ft) → E-Stopp signal pin landed; button PN TBD (Carling-style)	10A	#54	ORN/BLK/YEL	16
OUT8	20A	Cigarette Lighter / 12V Outlet	10A	#72	ORN/ORG	14
OUT9	8A	AMP Research Step Left	8A	#1	ORN/BLK	18
OUT10	8A	AMP Research Step Right	8A	#2	ORN/BLK/WHT	18
OUT11	8A	AMP Research Controller	2A	#3	ORN/BLK/BLK	22
OUT12	8A	Windshield Wiper Motor	6A	#49	PPL	18
OUT13	8A	park_tail group (Tail L/R + Park LF/RF)	1.6A	#75, 81, 83, 84	BRN/YEL, BRN/BLK, BRN/RED, BRN/BLU	22
OUT14	8A	Horn	5A	#48	ORN/ORG	16
OUT15	8A	backup group (Backup L/R + 3rd Brake + Backup Camera)	3.0–3.5A	#76, 89, 93, 97	BRN/VIO, BRN/ORG, BRN/RED, BLU/WHT	22
OUT16	8A	A/C Compressor Clutch	4A	#23	ORN/RED	20
OUT17	8A v4: legs #85a	LED Headlight Left (← floor dimmer COMMON) LOW / #85b HIGH (16 AWG) dimmer → lamp landed; #121 Dakota high-beam ind taps #85b	3.6A	#85	LT GRN	20
OUT18	8A v4: legs #86a	LED Headlight Right (← floor dimmer COMMON) LOW / #86b HIGH (16 AWG) paralleled at dimmer; dimmer PN/connector TBD	3.6A	#86	LT GRN/WHT	20
OUT19	8A	markers_clearance group (8 fixtures)	1.25–1.3A	#77, 78, 79, 87, 88, 90, 91, 92	BRN/ORG, BRN, BRN/WHT (x2), BRN (x2), BRN/YEL, BRN/VIO, BRN/BLK	22
OUT20	8A	Radio / Head Unit	3A	#31	ORN/BLK	22
OUT21	8A	Power Lock Actuator Left	3A	#38	BLK	20
OUT22	8A	Power Lock Actuator Right	3A	#47	BLK/WHT	20
OUT23	8A v4: powers Holley 558-499 T43 TCU (3A); CAN via #125 stub off trunk #62	Transmission Controller	3A	#58	ORN/YEL	22
OUT24	8A	Wideband Lambda Controller	3A	#64	ORN/BLK	22
OUT25	8A	interior_courtesy group (Dome, U-Dash, Footwell, U-Hood, Cargo)	1.9–2.5A	#67, 68, 69, 73, 74	BRN, BRN/WHT, BRN/BLK, BRN/RED, BRN/BLU	22
OUT26	8A	Washer Pump	2A	#50	ORN/BLK/RED	22
OUT27	8A v4: #119 Dakota	Turn Signal Left Front L-turn input = dash-side splice on #80	2A	#80 (+#119 tap)	LT BLU	22
OUT28	8A v4: #120 Dakota	Turn Signal Right Front R-turn input = dash-side splice on #82	2A	#82 (+#120 tap)	DK BLU	22
OUT29	8A v4: #124 Dakota switched-+12V backup splices OUT29 — verify vs Dakota internal jumper; #124 ORN/RED collides with #117 — reassign at mockup	Display / Dash (MoTeC C125)	1A	#65 (+#124 splice)	ORN/RED	22
OUT30	8A	USB Charging Port	1A	#70	ORN/YEL	22

POWER RAILS — CHANNEL TBD:
#INJ_PWR Injector +12V rail 16 AWG RED 5.0ft — feeds 8x EV6 injectors @ fuel rail — PDM CHANNEL TBD
#COIL_PWR Coil +12V rail 16 AWG RED 3.0ft — feeds 8x D510C @ DEL-Stributor bracket — PDM CHANNEL TBD
PDM-FED, CHANNEL NOT SPECIFIED IN CUT LIST v4:
#55 Transfer Case Ind · #56 Neutral Safety Sw · #57 Reverse Light Sw · #71 Dakota Digital (IGN rail tap, not a channel) · #94 Fuel Pump Rly · #95 iBooster Rly

NOT ON PDM — DIRECT WIRED
#59 Alternator 0 AWG ORN/VIO (220A source; capacity inconsistent across docs: 250A/220A/150A) #6 Starter Motor 0 AWG ORN (200A peak) #63 Battery Disconnect 0 AWG ORN/WHT (190A) #66 Fuel Pump 8 AWG ORN/BLU (35A, Aeromotive 16301 relay) #52 Bosch iBooster 8 AWG ORN (40A peak, dedicated relay) #32 Amplifier 8 AWG ORN/RED (30A, direct fused battery) — injectors/coils sw via M130 low-side

NOTE: stale harness_spec_latest.txt (2026-04-13) carries a different channel map (6x20A + 14x15A + 10x10A, E-Stopp ch7/8, QTP cutouts, keypad) — superseded by the 2026-04-05 authoritative plan shown above.

NUKE LTD / DESERT PERFORMANCE
1977 CHEVROLET K5 BLAZER — VIN CKR187F127263
LS3 6.2L / MoTeC M130 + PDM30
SHEET 3 OF 3 — 2026-06-09 — REV A
DERIVED FROM K5_cut_list_v4 + pdm30_channel_plan; lengths/land marks per calc engine